BUILD GUIDE - V1 BASIC - PARENT-FRIENDLY MODEL

Adjustable Therapy Bench

Simple Drill-Hole Version

24 x 10"	5" legs	4 rows	3/4"	No slots
TOP SURFACE	BASE HEIGHT	ADJUSTMENT	PLYWOOD	DRILL ONLY

Why this version exists

This V1 Basic version replaces routed slot channels with simple drilled holes. The goal is a bench that more parents can build with a circular saw or jigsaw, a drill, sandpaper, and common hardware.

What's inside

- Project overview and safety notes
- Materials, hardware, and tool list
- Exact cut list from one 24" x 24" plywood sheet
- One-page cut diagram
- Drill-hole adjustment layout
- Step-by-step assembly and inspection checklist

Project Overview

This bench supports floor sitting work and therapy-style activities for children who benefit from a firm, adjustable-height surface. The outside legs can be set to matching hole rows for a level bench or adjacent rows for a gentle tilt. This is a DIY woodworking plan, not medical advice - ask your child's therapist what height and angle are appropriate.

24 x 10"	5"	4 rows	12"	1 sheet
TOP	LEG HEIGHT	POSITIONS	OUTER BASE	24 X 24 PLY

Materials and Hardware

QTY	WOOD
1	24" x 24" sheet - 3/4" birch plywood
-	Optional scrap blocks for clamping and test drilling

QTY	HARDWARE
4	1/4"-20 bolts, 2-1/2" to 3" long
4	1/4"-20 star knobs
8	1/4" flat washers
4	1/4" lock nuts or nylon lock nuts
16	1" to 1-1/4" wood screws or brad nails

QTY	FINISH / SAFETY
-	80, 120, and 220 grit sandpaper
-	Water-based polyurethane, paint, or clear coat
-	Optional rubber feet or non-slip pads

TOOLS REQUIRED
Circular saw, jigsaw, miter saw, or handsaw
Drill/driver + 5/16" drill bit
Clamps, square, tape measure, pencil
Sander or sanding block

Parent-friendly build note

No router, no slot channels, and no complicated adjustment track. The adjustable feature is made from two columns of drilled holes in each outside leg.

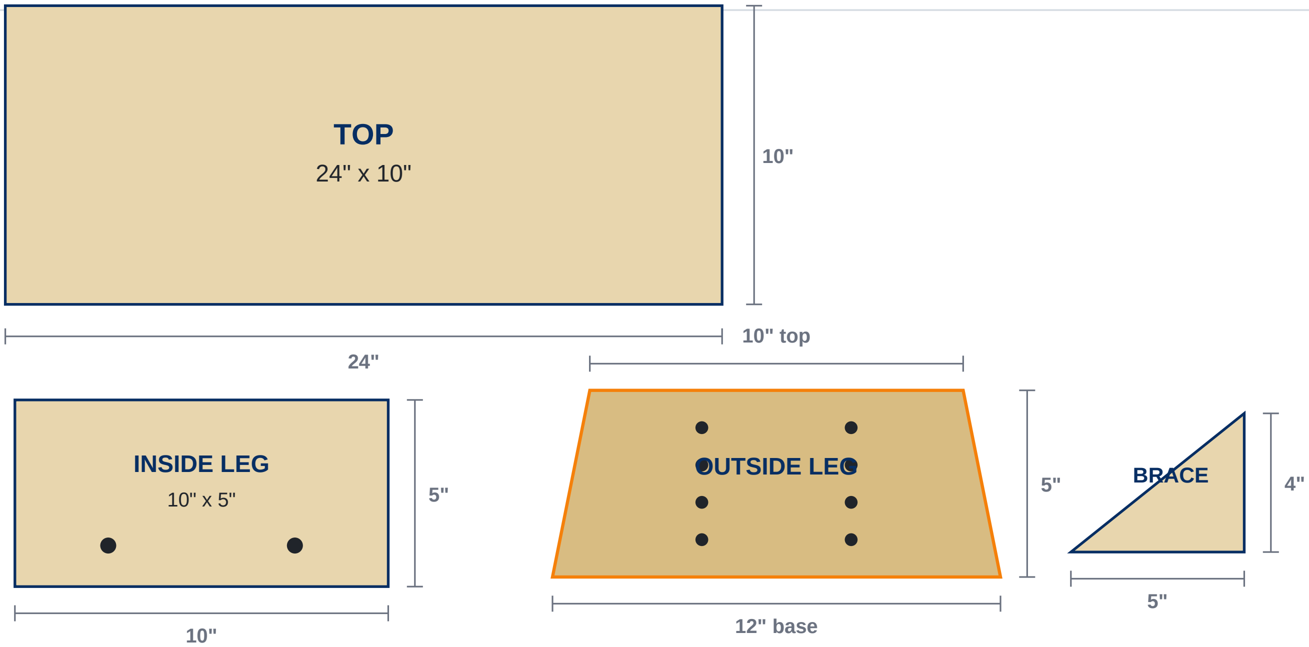
Cut List

PART	QTY	SIZE	NOTES
Top	1	24" x 10"	Main bench surface
Inside Leg	2	10" x 5"	Fixed under top; contains the captured bolts
Outside Leg	2	12" base x 5" tall; tapered to 10" top	Adjustable outer support; drill two columns of holes
Triangle Brace	4	5" x 4" right triangle	Installed on the outside of the top/inside-leg joint

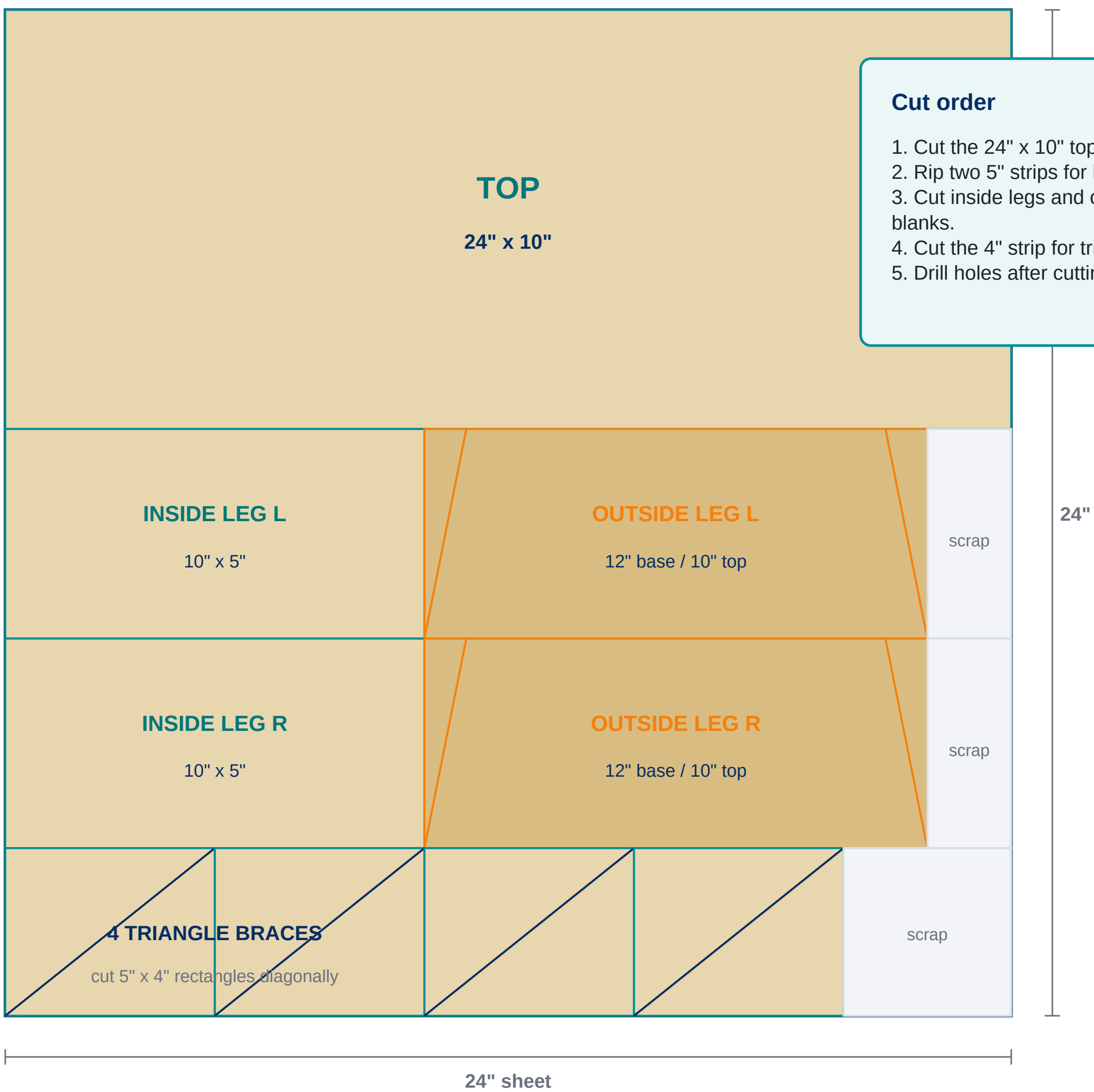
Taper note

Outside legs are symmetrical trapezoids: 12" wide at the base, 10" wide at the top, and 5" tall. The top is inset 1" on each side. Stack both outside-leg blanks and cut them together so they match.

Part Diagrams



One 24" x 24" Plywood Layout

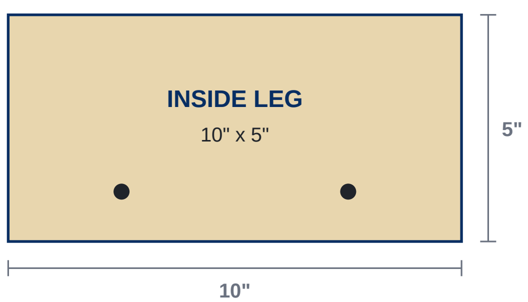


Cut order

1. Cut the 24" x 10" top first.
2. Rip two 5" strips for legs.
3. Cut inside legs and outside-leg blanks.
4. Cut the 4" strip for triangle braces.
5. Drill holes after cutting.

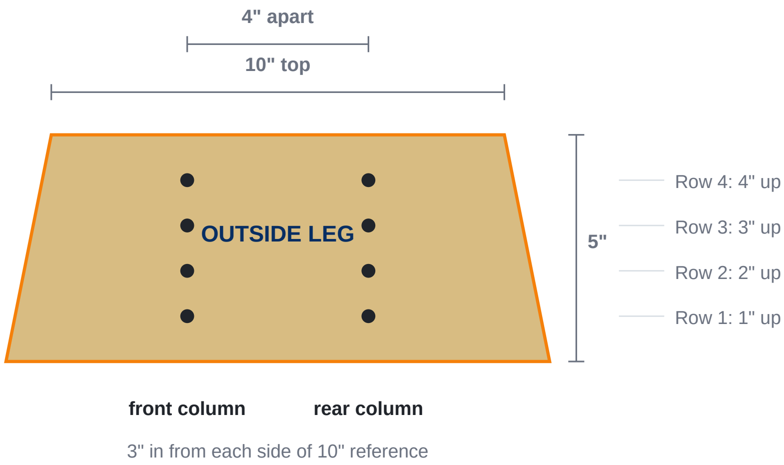
Drill-Hole Adjustment Layout

The inside leg has two fixed bolt holes near the bottom. The outside leg has matching front and rear columns with four height rows. Mark the columns 4 inches apart - 3 inches in from each side of the 10-inch inside-leg reference - so the outside leg slides onto the captured bolts cleanly.



Inside leg holes

Drill two holes near the bottom. These bolts become semi-permanent studs.



Recommended hole size

For 1/4" bolts, drill 5/16" holes. The slightly larger hole makes alignment easier and more forgiving for home builders.

Important

Mark and drill one outside leg first, then use it as the template for the second outside leg. Columns should be 4 inches apart, based on the 10-inch inside-leg reference.

Step-by-Step Assembly

1. **Sand and round all parts**
   Sand faces, break every edge, and round the top corners before final assembly.

2. **Attach inside legs to the top**
   Turn the top upside down. Glue and screw one 10" x 5" inside leg under each end of the top. Keep the legs square and centered across the 10" width.

3. **Add triangle braces**
   Attach two braces at each inside leg - one on each outside face of the top/leg joint. Use glue plus screws or brad nails.

4. **Capture the bolts in the inside legs**
   Insert two bolts through each inside leg from the inside toward the outside. Add washers and lock nuts so the bolts act like semi-permanent studs.

5. **Install outside legs**
   Slide each outside leg onto the exposed bolts at the desired hole row. Add washer and star knob on the outside. Tighten by hand.

6. **Check stability before use**
   Set the bench on a flat floor. Check for rocking, loose knobs, sharp edges, splinters, or cracks before every session.

Hardware and Adjustment

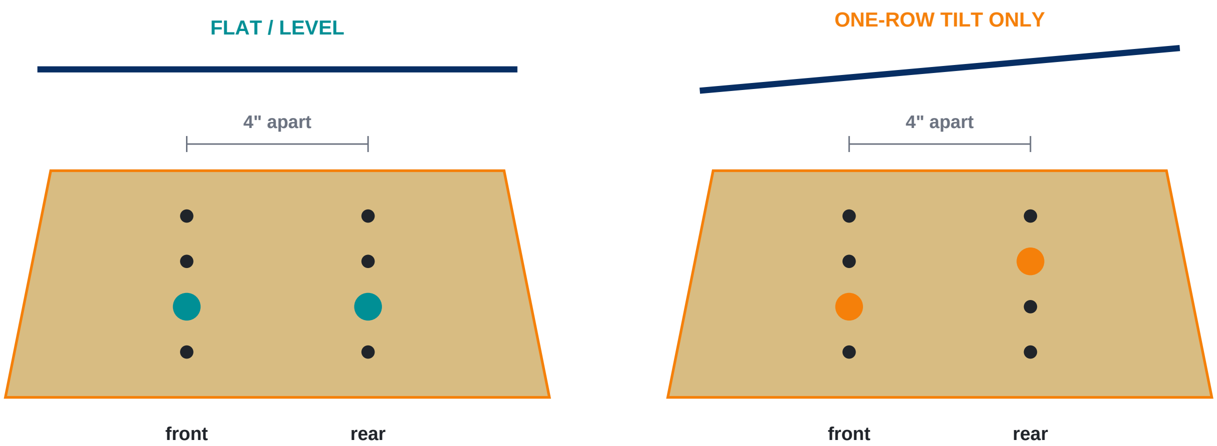
Hardware stack

INSIDE LEG > LOCK NUT > OUTSIDE LEG > WASHER > STAR KNOB

Bolts are captured through the inside leg first, then the outside leg slides over the exposed bolts. Tighten by hand with washers and star knobs.

One-side adjustment view

Each outside leg has two columns of holes. Use the same row in both columns for level height. For tilt, use only adjacent rows - a one-row difference between the front and rear columns.



Front Row 2 + rear Row 2 = flat

Front Row 3 + rear Row 3 = gentle tilt

Adjustment limit

The V1 Basic bench is designed for level positions or a gentle one-row tilt only. It cannot adjust more than one height either way; the bolts may not line up and the bench may become unstable.

Safety, Finishing, and Inspection

Edge Safety

Round every edge. Run your palm across the top, legs, braces, and holes. Anything sharp should be sanded again before use.

Splinter Check

After sanding, wipe with a damp cloth to raise the grain. Let dry, then sand again until smooth.

Hardware Check

Before every session, confirm all four knobs are tight and any tilt uses only adjacent rows.

Flat Floor Only

Use the bench on a flat surface. Do not use as a step stool or climbing object.

Regular Inspection

Stop using if you see cracks, loose screws, crushed holes, splinters, or unusual flex.

Therapist Guidance

Ask your child's therapist which height or tilt is appropriate. Do not force a position that feels unstable or uncomfortable.

Prototype reference photo

The photo shows the V1 Basic parts: 24" x 10" top, two inside legs, two wider-base outside legs with drilled adjustment holes, and four triangle braces.